chapter 1

Digital Systems and Binary numbers

Analog Sinyal: Sürekli değişim gösteren sinyaldir

Dijital Sinyal: Belirli aralıklarda değişim gösterir

→ Dijitalleşmenin en önemli amaçları analog sinyali özetleyerek gürültüden arıtma ve daha iyi işlenmesini sağlamaktır.

→ örnekleme yaparken ne kadar çok frekans yani kare alıp detaylı bakarsak o kadar iyi (doğru) sonuç alırım.

→ kuantalama veya örnekleme hatası çözünürlük artarsa azalır

Örnek

Binary Numbers: 0 and 1

- each digital represents a Power of 2, starting w 2^0

Hexadecimal: base 16

- Digits (0-9) and A-F (10-15)
- each digit represent a Power of 16, starting w 16^0
- to convert to/from binary: convert each hex digit to 4 binary digit

(A) ...when convert from binary/hexa/octal → multiply

Octal : base 8

- Digits : 0-7
- each digit represent a Power of 8
- to convert to/from binary : convert each octal to 3 binary

Binary addition method:

- Align numbers vertically and add corresponding bits, from right
- Carry over to the next bit if the sum is greater than or equal to (2)

Binary subtraction methods:

→ One's complement:

- negative numbers r represented by (tatla bir kalmasi)

26 → 11010 orijinal → 11010 → 1110 → 1111 = (15)
-11 → 10111 → +1 → 00100 → +1
(15) → 11110 → add carry

- if we have no carry we take the one' comp to the result (1 ucunu yoksa tersi alinir) sonuc negatif →

→ Two's Complement:

- 1010 → 10 → 1 → 2's complement → 1010
  0111 → -7 → +1000 → +1001 → result
  (3) → 1001 → 1|0011 → 0011 → (3)
  Ignore

- if we have no carry we take the 2's comp. to the final result sonuc negatif →

Chapter - 2 -

Logik Devre Temelleri

Sayısal Kodlama:

① BCD - binary Coded Decimal: ondalık sayının her hanesinin ikili olarak kodlanmasıdır.

$$(25)_{10} \rightarrow \underbrace{0010}_{2} + \underbrace{0101}_{5} = \boxed{0010\ 0101}$$

② üç fazlalık kodu: ikili sistemin üç fazlası alınarak oluşturulmuş. simetriktir.

0	0011
9	1100
1	0100
8	1011

simetrik

③ Aiken kodu: (0-4) arasındaki ilk beş sayının ikili kodlamaya eşit, (5-9) arasındaki beş sayının ise ilk beş sayının 1'e tümleyeni olduğu söylenebilir.

1	→ 0001
8	→ 1110

④ Bitişik kodlar (Gray): birbirini izleyen sayılara karşılık alınan ikili kod sözcükleri arasındaki uzaklık (hamming) 1 ise bitişik kod denir.

	00	01	11	10
00	0	1	2	3
01	9			4
11	8			5
10	7			6

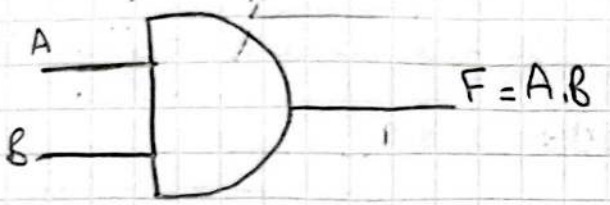
Gray farkı:

0	1	2	3
7	6	5	4
8	9	10	11
15	14	13	12

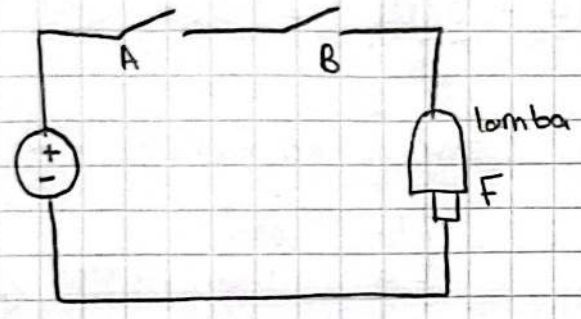
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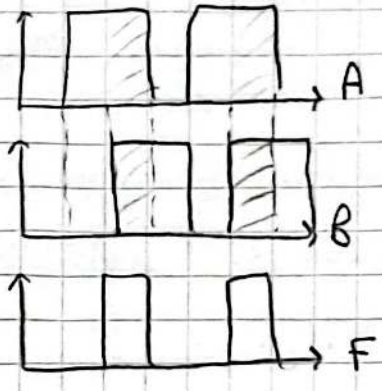
Boole Cebri:

① Ve işlemi AND:

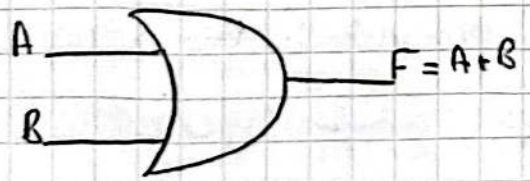


A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

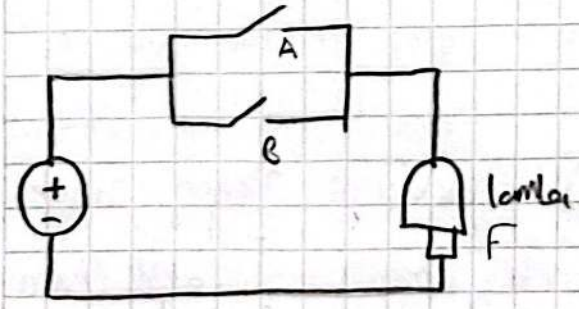




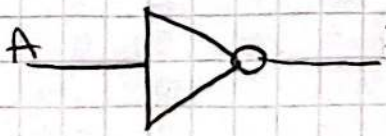
② Veya işlemi OR:



A	B	F
0	0	0
0	1	1
1	0	1
1	1	1



③ Tümleme işlemi (NOT):



A	Ā
0	1

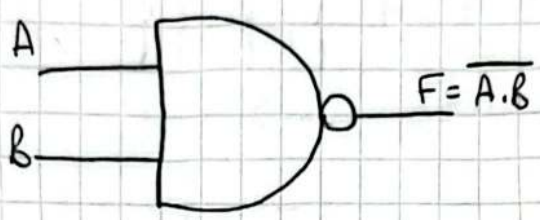
5V

GND



④ Tümleyen Ve işlemi (NAND):

Ve'nin değili:

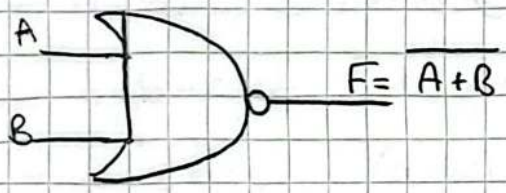


A	B	F = $\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

⑤ Tümleyen Veya işlemi (NOR):



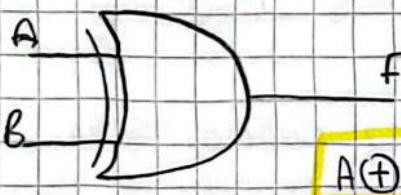
Veya'nın değili:



A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

⑥ Ya Da - Özel Veya - (XOR):

farklı olduklarında 1 aynı olduklarında 0



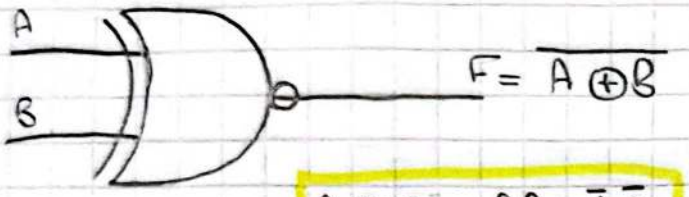
$A \oplus B = \bar{A}B + A\bar{B}$

A	b	F
0	0	0
0	1	1
1	0	1
1	1	0

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⑦ Tya Da (ancak ve ancak) işlem: (XNOR):

aynılarsa 1 farklılarsa 0



A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

$A \odot B = AB + \bar{A}\bar{B}$

- Soru -

A	1	1	0	1	0	1
B	1	1	0	0	0	1
F	1	1	1	0	1	1

XNOR
aynı → 1 farklı → 0

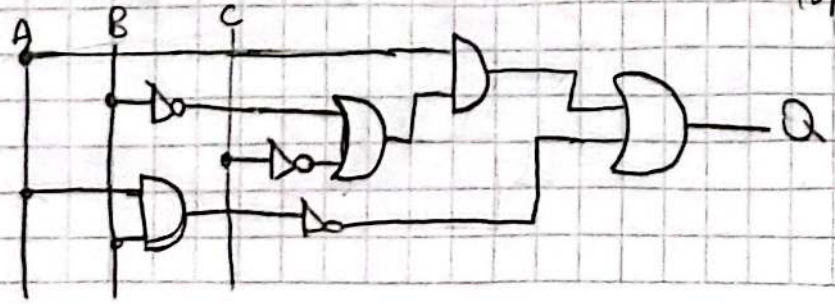
→ Basit kuramlar

$A + \bar{A} = 1$ $\quad 1 + A = 1$ $\quad 0 + A = A$

$A \cdot \bar{A} = 0$ $\quad 1 \cdot A = A$ $\quad 0 \cdot A = 0$

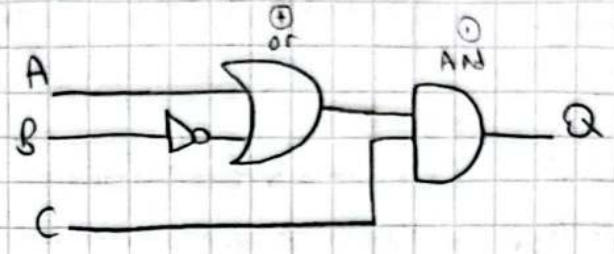
Soru: $Q = [A \cdot (\bar{B} + \bar{C})] + \overline{A \cdot B}$ ifadesinin lojik diyagramını çiziniz

çarpma → ve
toplama → veya



Kendi notlarım stoj.com

Soru: Aşağıdaki lojik diyagramın lojik ifadesini yazınız.



$$[(A+\bar{B})\cdot C] = Q$$

Teoremler:

→ $a+b+c = (a+b)+c$

→ $a\cdot b\cdot c = (a\cdot b)\cdot c = a\cdot(b\cdot c)$] birleşme

→ $a+(b\cdot c) = (a+b)\cdot(a+c)$

→ $a\cdot(b+c) = (a\cdot b)+(a\cdot c)$] Dağılma

→ $a + a\cdot b = a$

→ $a\cdot(a+b) = a$] yutma

Demorgan kuralı

$$\overline{(a+b+c+\dots)} = \bar{a}\cdot\bar{b}\cdot\bar{c}\cdot\dots$$

$$\overline{(a\cdot b\cdot c\cdot\dots)} = \bar{a}+\bar{b}+\bar{c}+\dots$$

Örnek: $Q = B\cdot C + B\cdot(C+A) + C\cdot(B+A)$ ifadesini boolean teoremleri ile sadeleştir.

$(B\cdot C)+(B\cdot C)+(B\cdot A)+(C\cdot B)+(C\cdot A)$

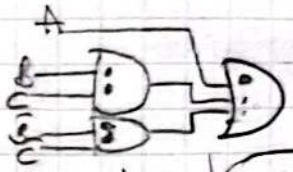
$(B\cdot C)+(B\cdot A)+(C\cdot A) = A\cdot(B+A)+B\cdot C$

$[A\cdot B\cdot(A+C)]$

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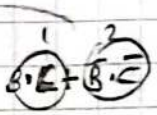
Circuit Optimization:

→ Literal cost (L)
→ Gate input cost (G)
→ Gate input cost with Not's (GN)



Literal Cost : (L) → Sadece bilinmeyenler ↓ 5 tane
the number of gates → inversion + the normal one

Gate input Cost (G) : ↦ 5 + 2 = 7



Gate input cost with Not's (GN) : Not kapılarını da say

7 + 2 = 9

Minterms:

Fonksiyonun doğru olduğu durumları ifade eder. (1)

$F(A,B,C) = \sum(0,2,5,7)$ → çıkışları 1 olan durumlar

→ çarpımların toplamı olarak hesaplanır (SOP)

Maxterms:

Fonksiyonun yalnış olduğu durumları ifade eder (0)

$F(A,B,C) = \prod(1,3,4,6)$ → çıkışları '0' olan durumlar

→ toplamların çarpımı olarak hesaplanır (POS) → 0

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Chapter -3-

Karnaugh Tablosu:

→ $2^2 = 4$ iki girişli

A \ B	B'	B
A'	A'B'	A'B
A	AB'	AB

$2^3 = 8$ 3 girişli

A \ BC	B'C'	B'C	BC	BC'
A'	A'B'C'	A'B'C	A'BC	A'BC'
A	AB'C'	A'B'C	ABC	ABC'

$2^4 = 16$

AB \ CD	C'D'	C'D	CD	CD'
A'B'	A'B'C'D'	A'B'C'D	A'B'CD	A'B'CD'
A'B	A'BC'D'	A'BC'D	A'BCD	A'BCD'
AB	AB'C'D'	AB'C'D	AB'CD	AB'CD'
AB'	ABC'D'	ABC'D	ABCD	ABCD'

! Sadeleştirme nasıl yapılır?

A \ B	B'	B
A'	0	1 → $f_2 = B$
A	1	1

$f_1 = A$

$Q = \bar{A}B + AB + A\bar{B}$ → $Q = f_1 + f_2 = A + B$

A+B

A \ BC	B'C'	B'C	BC	BC'
A'	0	1	1	0
A	0	0	1	1

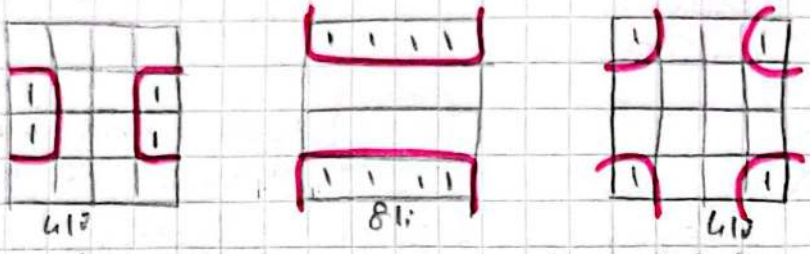
$Q = \bar{A}BC + \bar{A}B\bar{C} + AB\bar{C} + ABC$

$Q' = \bar{A}C + AB$

④ kırmızı olan gereksiz grup

$A'C + AB$

Yanlış gruplandırmalar:



→ Simetrik bir şekilde gruplandırılabilir

Quine-McCluskey yöntemi: 6 ve daha yüksek girişler için

- $m(0,1,2,5,7)$
- $\left.\begin{matrix} m_0 = 000 \\ m_1 = 001 \\ m_2 = 010 \end{matrix}\right\}$ 1 bit farkı
- 1 bit farkı $\left\{\begin{matrix} m_5 = 101 \\ m_7 = 111 \end{matrix}\right.$
- $(m_0, m_1, m_2)_{grup1}$, $(m_1, m_5, m_7)_{grup2}$
  (sadeleştir (1 bit))
- $(m_0, m_2)_1$, (m_1, m_5, m_7)
- $\boxed{Q = m_0 m_2 + m_1 m_5 m_7}$

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Toplama çıkarma devresi simülasyon

Chapter - 4 -

Combinational Circuit

Bu devrelerde çıkışlar yalnızca anlık girişlere bağlıdır ve devrenin geçmiş durumu ile ilgili değildir.

Bellek durumu taşımazlar, örnek; AND, OR, NOT gibi basit kapıların birleşmesiyle oluşan devrelere örnektir.

① Basit kapılar:

- AND, OR, NOT

- AND tüm girişler yüksek (1) çıkış yüksek, OR'da herhangi bir giriş yüksek dursa çıkış yüksek olur.

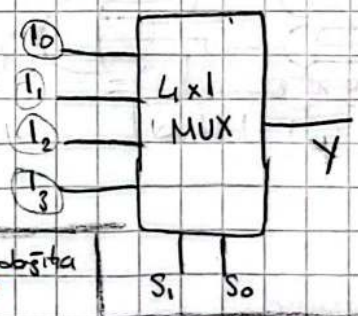
- iki girişli AND → çıkış = A ve B

② MUX (multiplexer): Seçici

- birden fazla giriş arasından seçim yapmak için kullanılır. Kontrol sinyali hangi girişin çıkışa ileteceğini söyler

- iki girişli bir MUX:

$$\text{çıkış} = \begin{pmatrix} \text{kontrol} = 0 \rightarrow \text{giriş} = 1 \\ \text{kontrol} = 1 \rightarrow \text{giriş} = 2 \end{pmatrix}$$



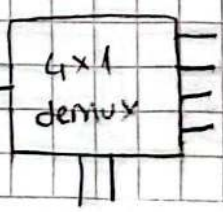
S_0	S_1	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

③ DEMUX (demultiplexer): dağıtıcı

- Demux tek giriş, fazla çıkış. Kontrol sinyali hangi çıkışın girişe bağlanacağını belirler.

- tek girişi iki farklı çıkışa gösteren deMUX:

$$\text{Giriş} = \begin{pmatrix} \text{kontrol} = 0 \rightarrow \text{çıkış} = 1 \\ \text{kontrol} = 1 \rightarrow \text{çıkış} = 2 \end{pmatrix}$$



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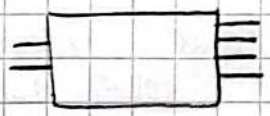
4 Encoder ve Decoder (Kodlayıcılar ve Çözücüler)

Encoder: birden fazla girişi daha az girişe kodlar

Decoder: Kodlanmış bilgiyi orijinal girişlere dönüştürür.
low active → inverter high active → normal olan

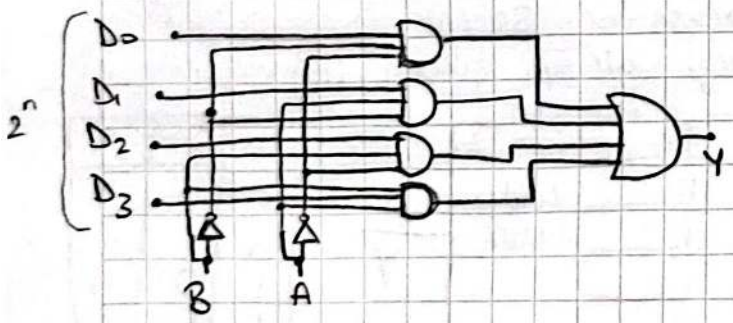
Örneğin: 4-2 kodlayıcı → 4 girişli bir veri setini 2 girişli bir veri setine kodlar.

2-4 çözücü → 2 girişli bir veri setini 4 farklı çıkışa dönüştüren bir devredir.
(2 giriş - 4 çıkış)

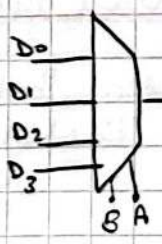


ÖRNEKLER:

1) 4 to 1 MUX

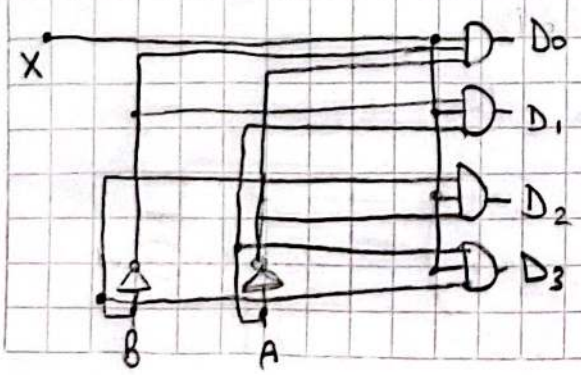


$D_0 = \bar{A} \cdot \bar{B}$
$D_1 = \bar{B} \cdot A$
$D_2 = B \cdot \bar{A}$
$D_3 = A \cdot B$

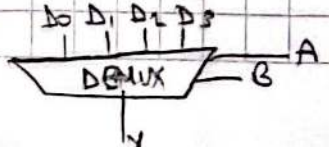


B	A	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

2) 1 to 4 DEMUX



B	A	D_0	D_1	D_2	D_3
0	0	X	0	0	0
0	1	0	X	0	0
1	0	0	0	X	0
1	1	0	0	0	X

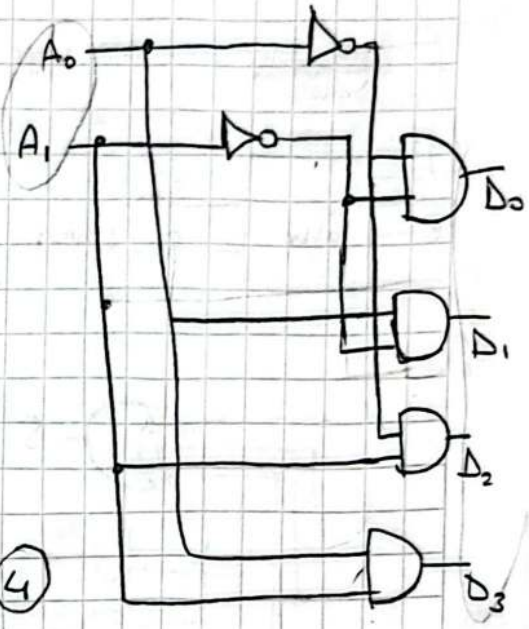


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Decoder Examples:

2 to 4

A_1	A_0	D_0	D_1	D_2	D_3
0	0	[1]	0	0	0
0	1	0	[1]	0	0
1	0	0	0	[1]	0
1	1	0	0	0	[1]



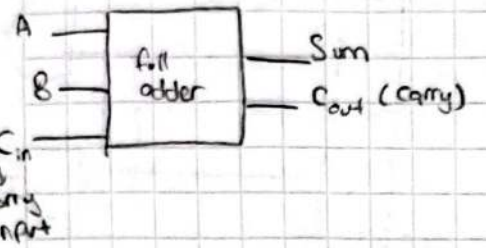
$n \rightarrow m \quad m = 2^n \rightarrow 2^2 = (4)$

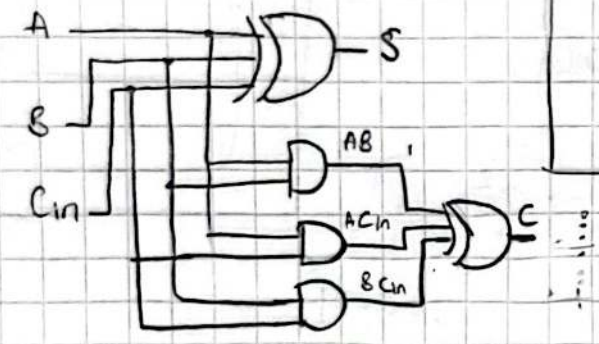
Encoding Example

Arithmetic addition - Subtraction circuits:

① Arithmetic addition circuit:

full-adder : compute longer numbers

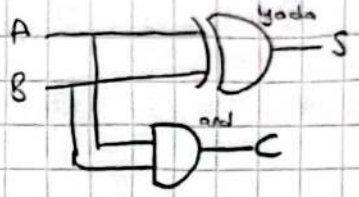




half-adder : compute single digit numbers



A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

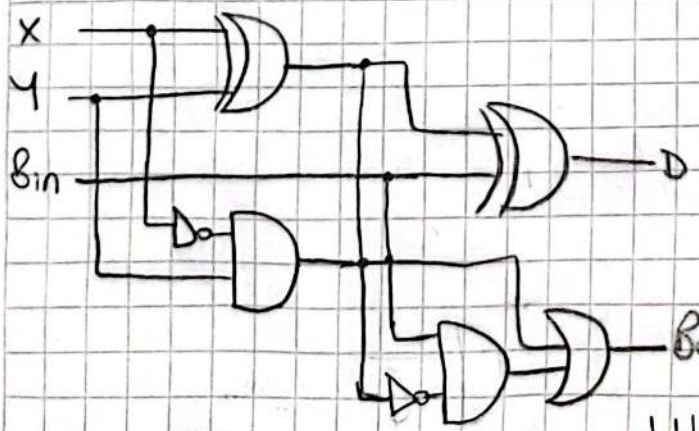


A	B	Cin	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

② Arithmetic Subtraction Circuit:

$D = X - Y - B_{in}$

full-subtractor:



X	Y	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

14 − 5 = 9

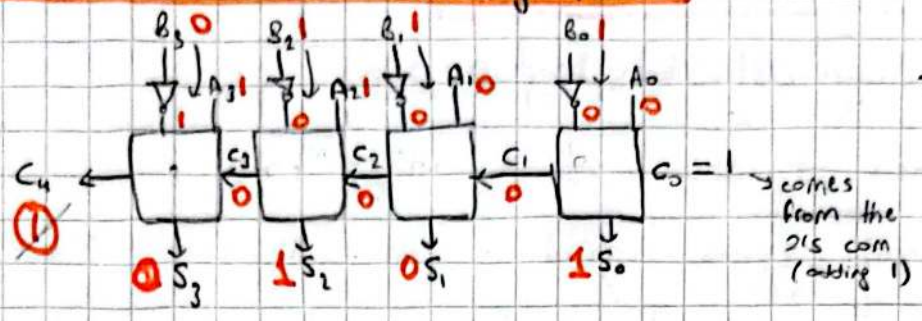
1110 − 0101 → two's comp. (1011) → 1110 − 1011

(1) 1001 → 9

1001

öder → devre görsel + input-çıkış + açıklama

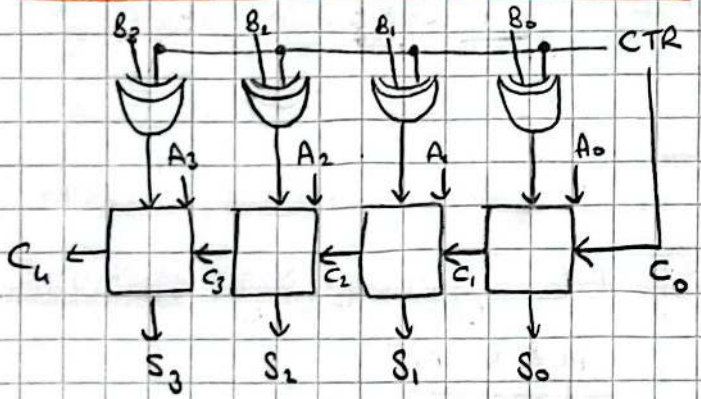
4 bit Subtractor using Adder :



12 → 1100 A
-7 → -0111 B
↓ 2's comp.
1100
+1001
X0101

Solution = 0101 = 5

4 bit Adder / Subtractor :



0101 = A
0001 = B
111
0101
+ 0111
X 0100 = 4

① 12 + 7 (Sum)
100
1100 A
+0111 B
10011 → 19
16 2 1

CTR = 0

Adder
7 + 3
0111
+ 0011
1010

② 12 − 7 (subtract)
1100 − 0111 → 2's → 0000 1100 +1001 X0101 → 0101 = 5

CTR = 1

Subtractor

③ A < B
10 − 12
1010 − 1100 → 1010 +0100 = 1110 = ??? no carry

Complicated Case

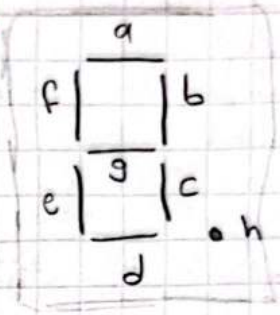
0011

3-state buffer < high / low

Chapter - 5-

ALU (orithmetic logic unit)

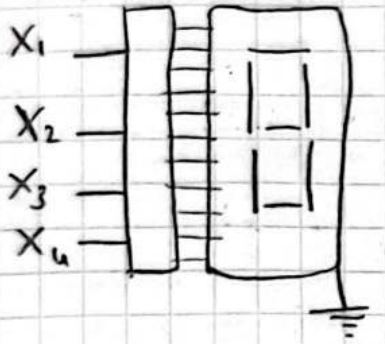
Seven Segment Display



1 sayısı → bc

2 ⊇ sayısı (2) → abged

gibi

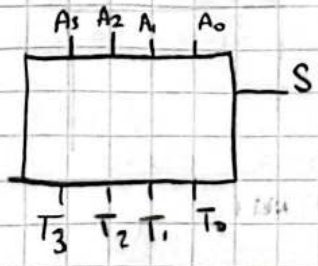


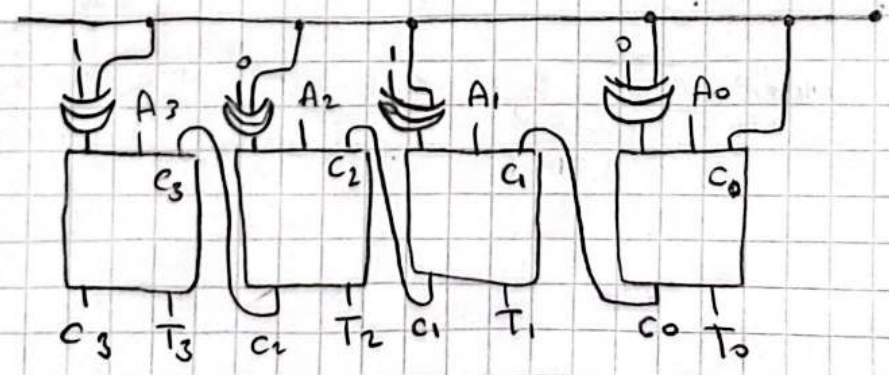
Örnek:

bu devreyi tam toplayıcılarla yapınız, XOR kullanabilirsin

$S_0 \rightarrow T = A+2$

$S_1 \rightarrow T = A-2$

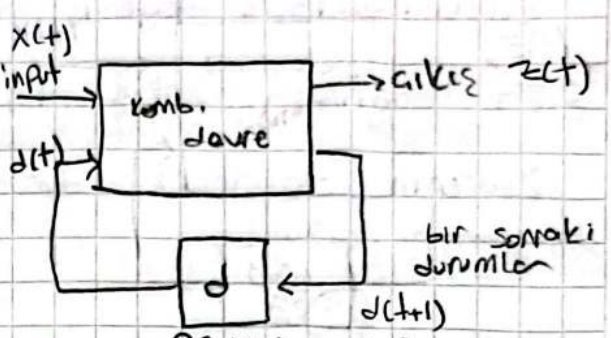




Chapter - 6 -

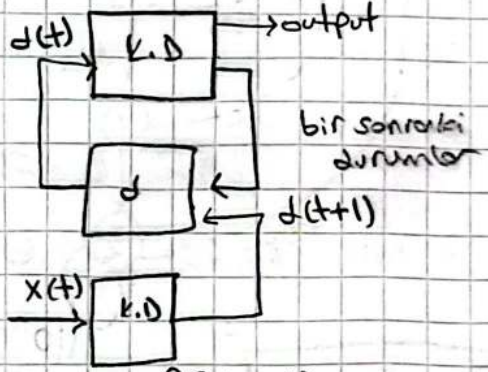
Sequential Circuits

- Mealy makinalarında t anındaki çıkış değeri o andaki durumlara ve girişlere bağlıdır.



→ $z(t) = f(d(t), x(t)) = d(t+1)$

- Moore makinalarında ise t anındaki çıkış değeri yalnızca o andaki durumlara bağlıdır.

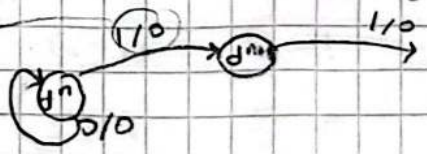


→ $z(t) = f(d(t))$

→ $d(t+1) = f(d(t), x(t))$

Durum diyagramı:

Ardışıl devrenin durumlarını ve durumlar arası geçişi gösteren graftır



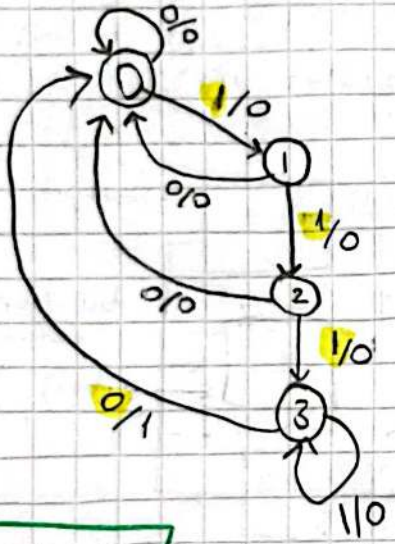
geçiş koşulu / çıkış değeri

Örnek:

Şifre devresi tasarımı

Şifrenin lojik değerde 1110 olduğu varsayılmaktadır. Şifre doğruysa çıkış 1, değilse 0 olmalı

Devrenin olabilecek durumlarını belirleyin ve durum diyagramını çizinit.

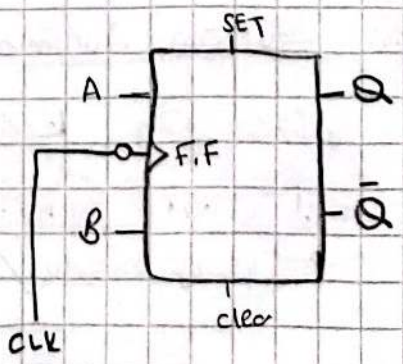


1110 → çıkış 1 √√

Flip-floplar:

Flip floplar 1 bitlik saklama birimleridir.

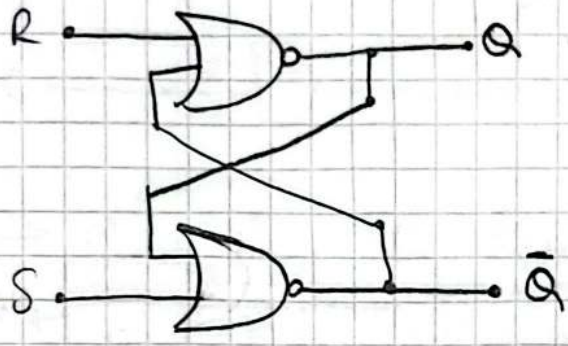
Saklayıcı, sayıcı, bellek vb. birimler oluşturabilir



(Set - Reset Latch):

- NOR
- NAND

- basic digital circuit element that can store one bit. 1 or 0.
- also known as a flipflop or bistable multivibrator



Reset → output = 0 = Q

Set → output = 1 = Q

Case 1:

S=0 R=1 Q=0 $\bar{Q}$=1

S=0 R=0 Q=0 $\bar{Q}$=1 memory

Case 2:

S=1 R=0 Q=1 $\bar{Q}$=0

S=0 R=0 Q=1 $\bar{Q}$=0 memory

Case 3 NOT USED! (1,1) → for NOR (0,0) → for Nand

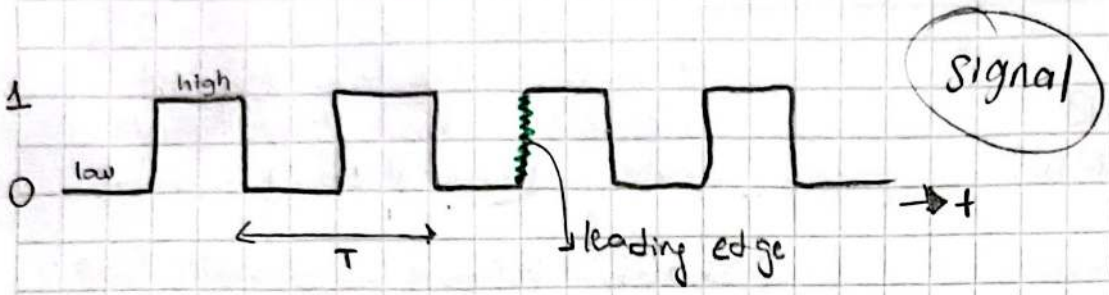
S=1 R=1 Q=0 $\bar{Q}$=0 (X)

S=0 R=0 Q=0 $\bar{Q}$=1 ???

S=0 R=0 Q=1 $\bar{Q}$=0 ???

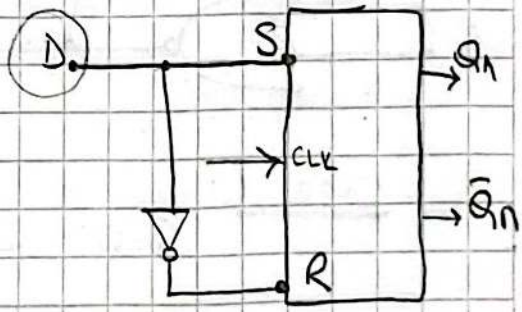
Scanned with CamScanner

what is clock ?



D flip flop (data) or delay

CLK	S	R	Q_{n+1}
0	x	x	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	invalid



truth table

clk	D	Q_{n+1}
0	x	Q_n
1	0	0
1	1	1

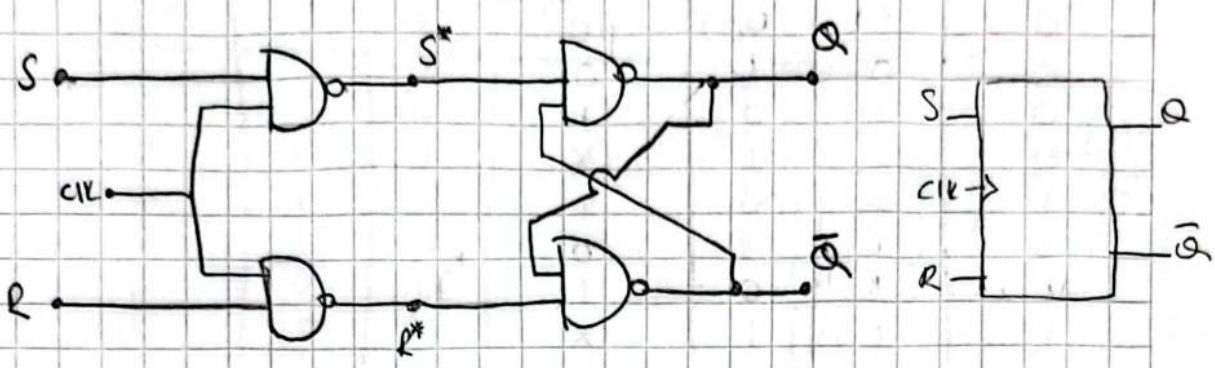
characteristic table

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

$Q_{n+1} = D = S$

$Q_n \rightarrow Q_{n+1}$ 10 11

SR flip-flop:



→

S^*	R^*	Q	$\bar{Q}$
0	0	not used (Nand) (in Nor (111))	
0	1	1	0
1	0	0	1
1	1	memory	

* $S^* = \overline{(S \cdot CLK)} = \bar{S} + \overline{CLK}$ $\quad R^* = \overline{(R \cdot CLK)} = \bar{R} + \overline{CLK}$

→

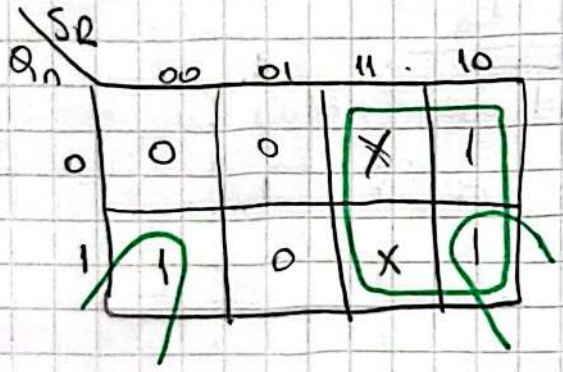
CLK	S	R	Q	$\bar{Q}$
0	x	x	memory	
1	0	1	0	1
1	1	0	1	0
1	1	1	not used	
1	0	0	memory	

Scanned with CamScanner

SR flip-flop

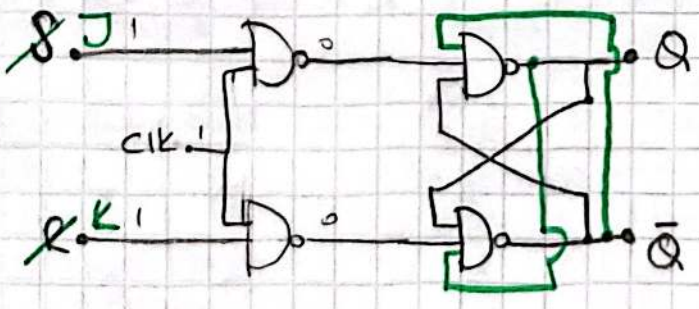
Characteristic table:

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X



$Q_{n+1} = S + Q_n\bar{R}$

JK flip flop



CLK	J	K	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	$\bar{Q}_n$ (toggle)

Scanned with CamScanner

characteristic table (JK):

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0'=1
1	0	0	1 memory
1	0	1	0
1	1	0	1
1	1	1	1'=0

Q_n \ JK	00	01	11	10
0	0	0	1	1
1	1	0	0	1

$$Q_{n+1} = \overline{Q_n}J + Q_nK$$

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

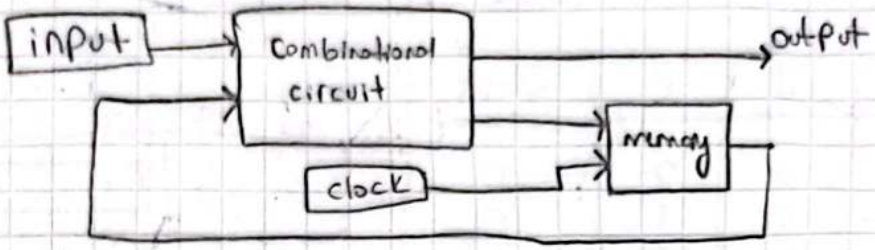
Q_n \ Q_{n+1}	0	1
0	0	1
1	X	X

$$J = Q_{n+1}$$

Q_n \ Q_{n+1}	0	1
0	X	X
1	1	0

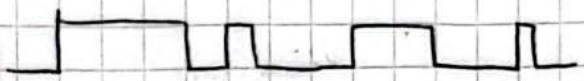
$$K = \overline{Q_{n+1}}$$

→ Triggering methods in flip-flops



→ Difference between Latch and flip-flop

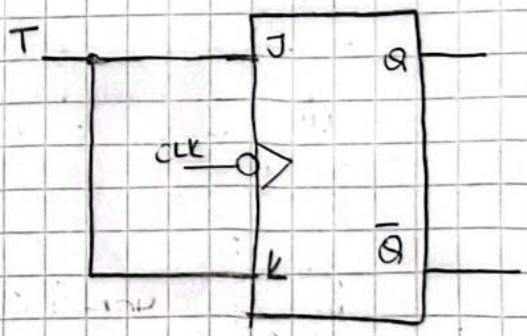
→ there is enable in latch



→ there is clock in flip flop



T flip-flop (toggle)



CLK	T	Qn+1
0	X	Qn (memory)
1	0	Qn (memory)
1	1	$\bar{Q}_n$ (toggle)

* ch. table

Qn	T	Qn+1
0	0	0
0	1	1 → 0'
1	0	1
1	1	0 → 1'

* Excitation table

Qn	Qn+1	T
0	0	0
0	1	1
1	0	1
1	1	0

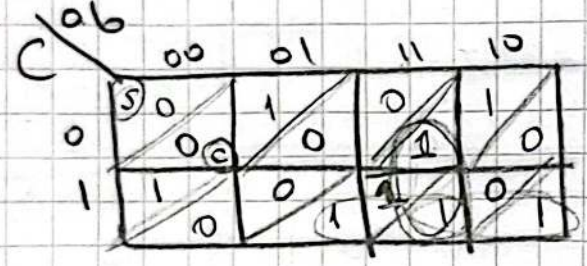
$Q_{n+1} = Q_n \oplus T$

XOR

T=0 → memory T=1 → $\bar{Q}_n$ → terim yapar

Full-adder

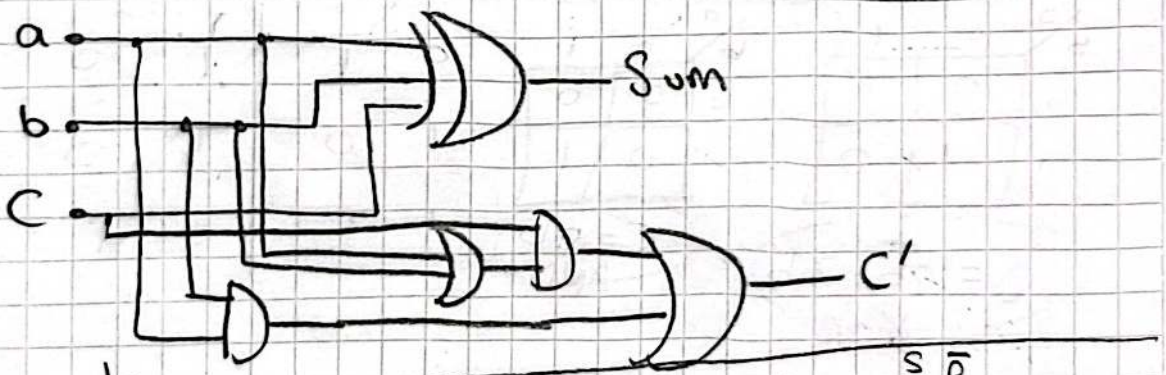
C	a	b	S	C'
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

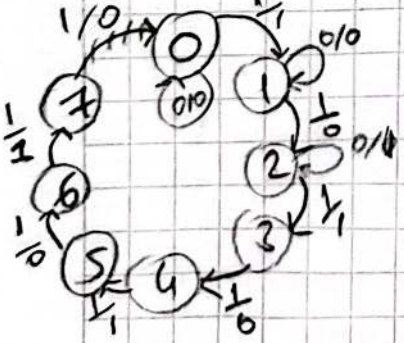


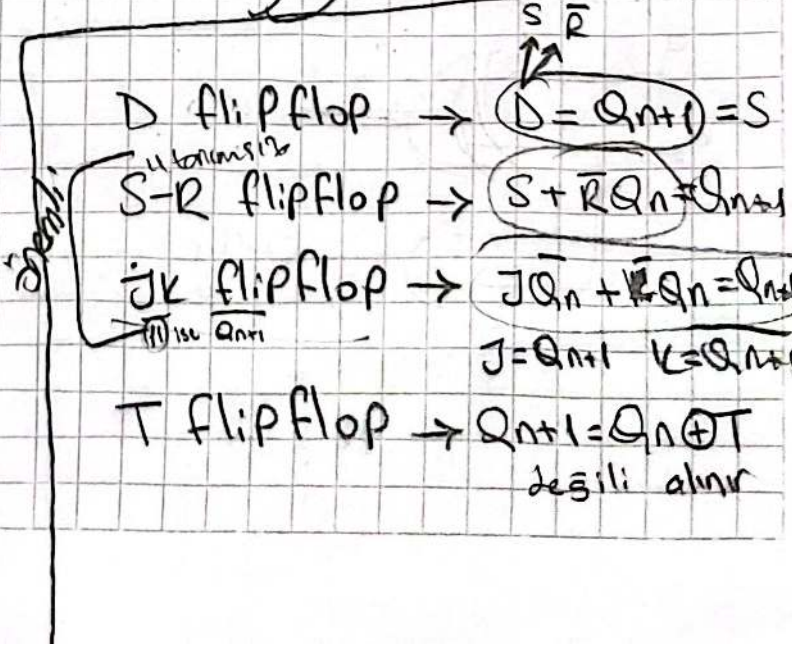
$S = c \oplus a \oplus b$ (XOR)

$c' = ab + cb + ac$

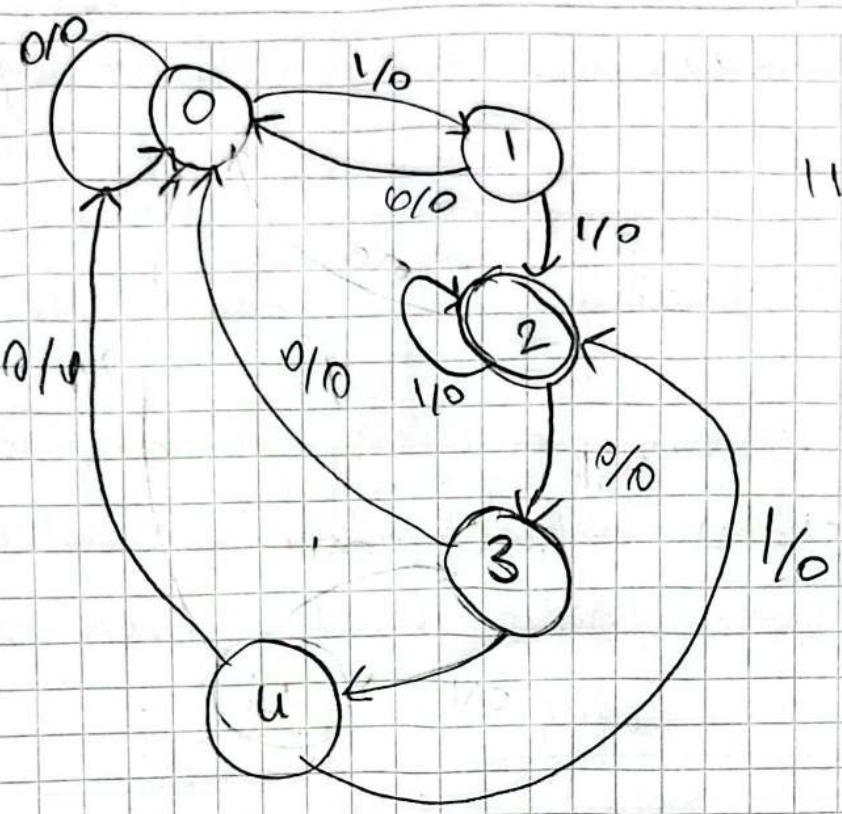
$c' = ab + c(a+b)$





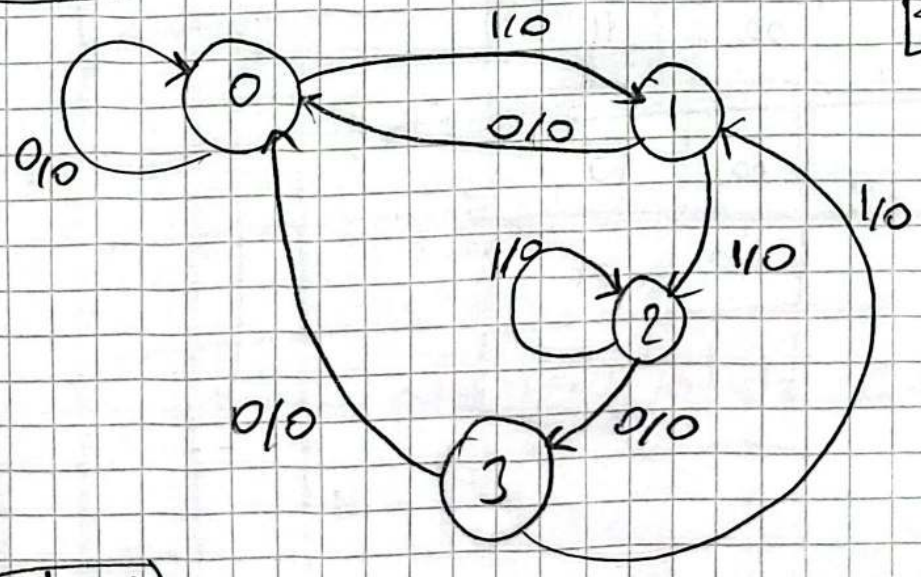


1101101



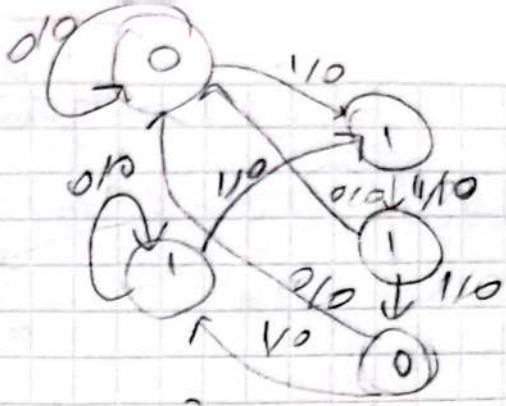
1101101

2 bit



1101

hafta 12 #



mealy
moore → output state

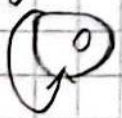
Clock

mealy gösteriminde çıkış değişebilir.

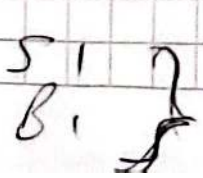
moor : sonuç yani çıkış aynı olduğu için
f(state) = output altına yazabiliyor
ama bake çıkış farklı olursa 2. durum düşün.

Durum Q	Durum I		Çıkış
	X=0	X=1	
11	00	10	
01	00	11	
10	00	10	
00	00	01	

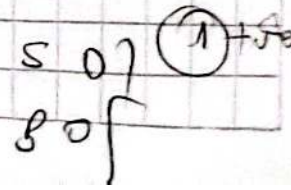
Example 3: State Diagram



S^0 01 → değişimi tuşa devre
B^1



0 → tuşa 1.1 sn



stride

1 = 5V
0 = GND

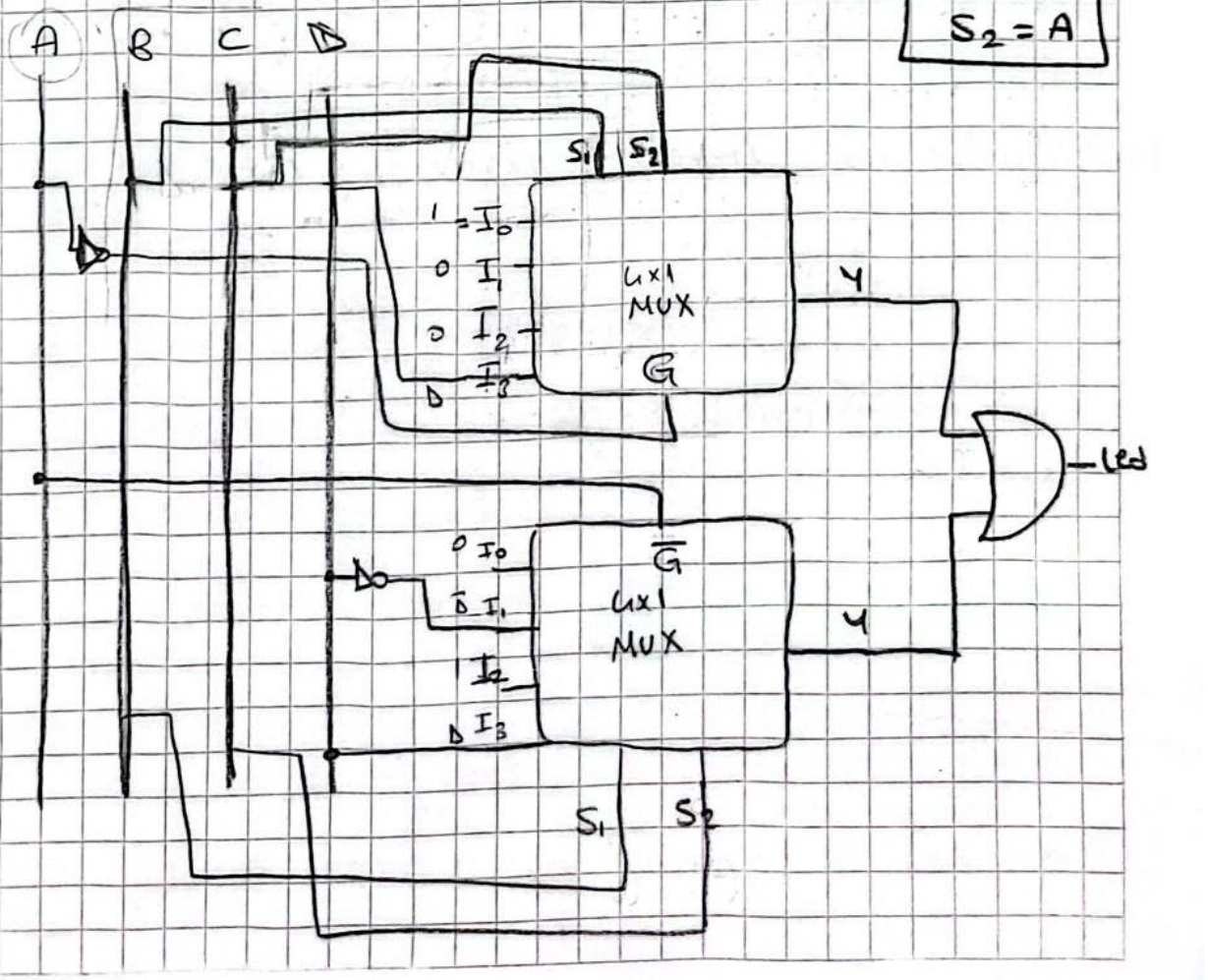
G (B) (A)

A	B	C	D	F	
0	0	0	0	1	1 = I_0
0	0	0	1	1	
0	0	1	0	0	0 = I_1
0	0	1	1	0	
0	1	0	0	0	0 = I_2
0	1	0	1	0	
0	1	1	0	0	D = I_3
0	1	1	1	0	
1	0	0	0	0	0 = I_0
1	0	0	1	0	
1	0	1	0	1	$\bar{D}$ = I_1
1	0	1	1	0	
1	1	0	0	1	1 = I_2
1	1	0	1	1	
1	1	1	0	0	D = I_3
1	1	1	1	1	

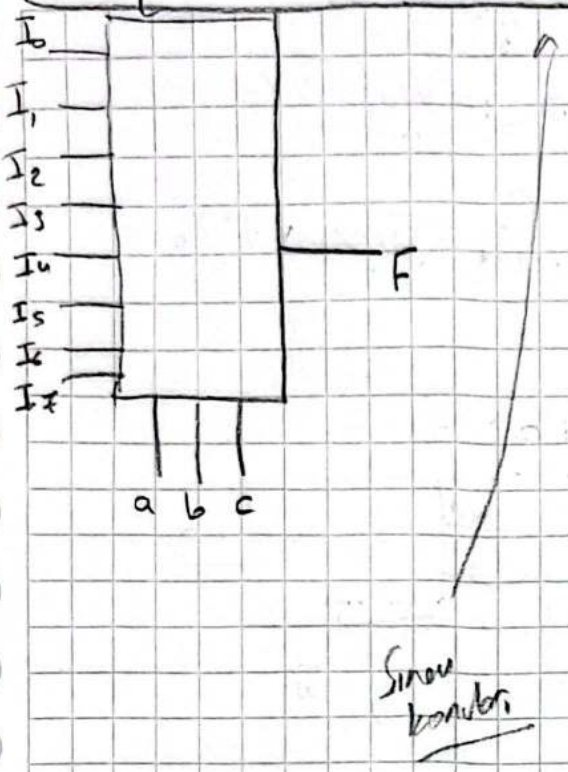
0 1 1 0 0 → D

desired

$S_1 = B$
$S_2 = A$



7-combo devre | 1 minterm maxterm | 3 karnaug
8-cost | 2 sayısal gösterimler | 4 lojik sentez
6 mux decoder | 5 " analiz



a	b	c	d	F	
0	0	0	0	1	I_0
0	0	0	1	1	
0	0	1	0	1	I_1
0	0	1	1	0	
0	1	0	0	1	I_2
0	1	0	1	1	
0	1	1	0	1	I_3
0	1	1	1	1	
1	0	0	0	1	I_4
1	0	0	1	0	
1	0	1	0	1	I_5
1	0	1	1	0	
1	1	0	0	1	I_6
1	1	0	1	1	
1	1	1	0	1	I_7
1	1	1	1	1	

Sequential Circuits

→ depolama elemanı (memory)

+ outputun çıkmasını sağlayan son adım (sistem) saklar.

Adım

→ Storage element → flip-flop

→ mealy → önceki durum (geçmiş)

→ moore → sadece şimdi

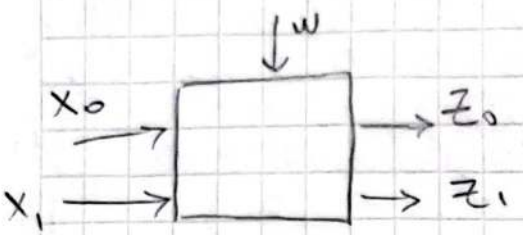
→ gecikme zamanını kullanıp eski değere erişmek amaçlanır.

String state (continued)

→ önemli eleman (clock) → (δ) sonra bunağa değiştir.

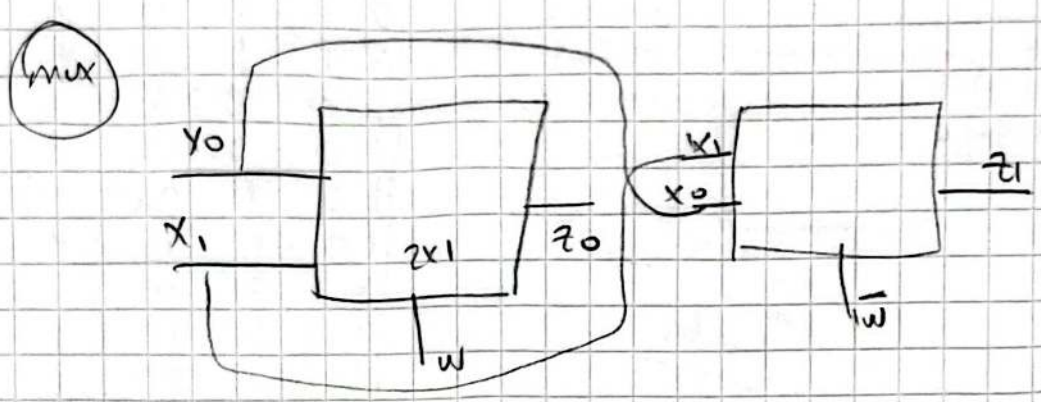
hafta 10

Soru 4



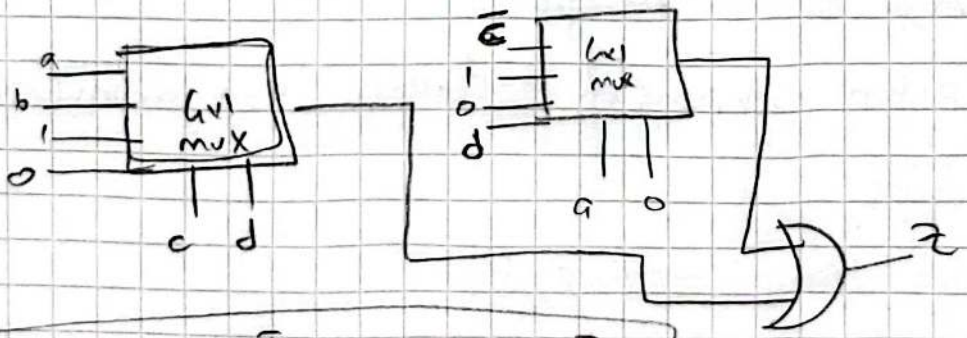
$f_{z_0} = w x_1 + \bar{w} x_0$

$f_{z_1} = \bar{w} x_1 + w x_0$



Soru -17

2 tane mux ve bir veya kapısı



$f_{mux_1} = \bar{c}\bar{d}a + \bar{c}db + c\bar{d}1 + cd0$

$f_{mux_2} = \bar{a}\bar{b}\bar{c} + \bar{a}b1 + a\bar{b}0 + abd$

→ a b c yi seçme girişleri olarak deneyi 8x1 lik mux yapın

$2^3 = 8$

3 = seçme